

Bose (-) 和 Fermi (+) 的平衡态分布 $n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i} \pm 1}$ 约束: $N = \sum_i \frac{g_i}{e^{\alpha + \beta \epsilon_i} \pm 1}$, $\bar{E} = \sum_i \frac{g_i \epsilon_i}{e^{\alpha + \beta \epsilon_i} \pm 1}$

引入 Bose 和 Fermi 系统的特性函数

$$\Phi(\alpha, \beta, \gamma) = \pm \sum_i g_i \ln(1 \pm e^{-\alpha - \beta \epsilon_i})$$

$$N = \sum_i \frac{g_i}{e^{\alpha + \beta \epsilon_i} \pm 1} = \sum_i \frac{g_i e^{-\alpha - \beta \epsilon_i}}{1 \pm e^{-\alpha - \beta \epsilon_i}} = -\left(\frac{\partial \Phi}{\partial \alpha}\right)_{\beta, \gamma}, \quad \bar{E} = \sum_i \frac{g_i \epsilon_i}{e^{\alpha + \beta \epsilon_i} \pm 1} = \sum_i \frac{g_i \epsilon_i e^{-\alpha - \beta \epsilon_i}}{1 \pm e^{-\alpha - \beta \epsilon_i}} = -\left(\frac{\partial \Phi}{\partial \beta}\right)_{\alpha, \gamma}$$

注意是用 N 约束 α , 所以 $\alpha = \alpha(\beta, \gamma, N)$

$$\bar{E} = \bar{E}(\beta, \gamma, N)$$

$$Y_k = \sum_i n_i \frac{\partial \epsilon_i}{\partial y_k} = \sum_i \frac{g_i e^{-\alpha - \beta \epsilon_i}}{1 \pm e^{-\alpha - \beta \epsilon_i}} \cdot \frac{\partial \epsilon_i}{\partial y_k} = -\frac{1}{\beta} \frac{\partial \Phi}{\partial y_k} \Rightarrow P = \frac{1}{\beta} \frac{\partial \Phi}{\partial V}, \quad T = T(\beta, V, N)$$

$$\epsilon_i = \epsilon_i(y_1, y_2, \dots, y_k, \dots, y_L)$$

$$dQ = d\bar{E} - \sum_k Y_k dy_k \quad d\Phi = \frac{\partial \Phi}{\partial \beta} d\beta + \frac{\partial \Phi}{\partial \alpha} d\alpha + \sum_k \frac{\partial \Phi}{\partial y_k} dy_k$$

$$= -d\left(\frac{\partial \Phi}{\partial \beta}\right) + \frac{1}{\beta} \sum_k \frac{\partial \Phi}{\partial y_k} dy_k$$

$$= -d\left(\frac{\partial \Phi}{\partial \beta}\right) + \frac{1}{\beta} \left(d\Phi - \frac{\partial \Phi}{\partial \beta} d\beta - \frac{\partial \Phi}{\partial \alpha} d\alpha\right)$$

$$= \frac{1}{\beta} d\left(\Phi - \beta \frac{\partial \Phi}{\partial \beta}\right) - \frac{1}{\beta} \frac{\partial \Phi}{\partial \alpha} d\alpha = \frac{1}{\beta} d\left(\Phi - \beta \frac{\partial \Phi}{\partial \beta} - \alpha \frac{\partial \Phi}{\partial \alpha}\right) + \frac{1}{\beta} \alpha d\left(\frac{\partial \Phi}{\partial \alpha}\right)$$

$$= \frac{1}{\beta} d\left(\Phi - \beta \frac{\partial \Phi}{\partial \beta} - \alpha \frac{\partial \Phi}{\partial \alpha}\right)$$

$$\Rightarrow \beta = \frac{1}{kT}, \quad S = k\left(\Phi - \beta \frac{\partial \Phi}{\partial \beta} - \alpha \frac{\partial \Phi}{\partial \alpha}\right) + S'$$

$$W_B\{n_i\} = \prod_i \binom{n_i + g_i - 1}{g_i - 1}, \quad W_F\{n_i\} = \prod_i \binom{g_i}{n_i}$$

$$\ln W_B\{n_i\} \approx \sum_i \left[(n_i + g_i - 1) \ln(n_i + g_i - 1) - n_i \ln n_i - (g_i - 1) \ln(g_i - 1) \right]$$

$$= \sum_i n_i \left(\frac{g_i - 1}{n_i} + 1 \right) + (g_i - 1) \ln \left(1 + \frac{n_i}{g_i - 1} \right)$$

$$\ln W_F\{n_i\} \approx \sum_i \left[g_i \ln g_i - n_i \ln n_i - (g_i - n_i) \ln(g_i - n_i) \right]$$

$$= \sum_i n_i \ln \left(\frac{g_i}{n_i} - 1 \right) + g_i \ln \left(1 - \frac{n_i}{g_i} \right)$$

$$\ln W_F\{n_i\} \approx \sum_i \left[n_i \ln \left(\frac{g_i}{n_i} + 1 \right) + g_i \ln \left(1 + \frac{n_i}{g_i} \right) \right]$$

$$= \sum_i \left[n_i (\alpha + \beta \epsilon_i) + g_i \ln \left(1 \mp \frac{1}{e^{\alpha + \beta \epsilon_i} \pm 1} \right) \right]$$

$$= \sum_i n_i (\alpha + \beta \epsilon_i) \pm g_i \ln \left(\frac{e^{\alpha + \beta \epsilon_i} \pm 1}{e^{\alpha + \beta \epsilon_i}} \right)$$

$$= \sum_i \left[n_i (\alpha + \beta \epsilon_i) \pm g_i \ln(1 \pm e^{-\alpha - \beta \epsilon_i}) \right]$$

$$= \Phi - \alpha \left(\frac{\partial \Phi}{\partial \alpha} \right) - \beta \frac{\partial \Phi}{\partial \beta} \Rightarrow S = \ln W_F\{n_i\} \xrightarrow{\frac{1}{k}} \frac{1}{k} \left[\Phi - \alpha \frac{\partial \Phi}{\partial \alpha} - \beta \frac{\partial \Phi}{\partial \beta} \right]$$

$$= k \left[\Phi - \alpha \frac{\partial \Phi}{\partial \alpha} - \beta \frac{\partial \Phi}{\partial \beta} \right] \Rightarrow S' = 0$$

对于开放系统, $d\bar{E} = d\bar{E} - \sum_k Y_k dy_k = \mu dN$

$$d\bar{E} = \sum_k Y_k dy_k = T dS + \mu dN$$

$$d\bar{E} - \sum_k Y_k dy_k = T dS + \mu dN = \frac{1}{\beta} d\left(\Phi - \beta \frac{\partial \Phi}{\partial \beta} - \alpha \frac{\partial \Phi}{\partial \alpha}\right) + \left[\frac{\alpha}{\beta} d\left(\frac{\partial \Phi}{\partial \alpha}\right)\right] = -\frac{\alpha}{\beta} dN \Rightarrow \alpha = -\mu\beta = -\frac{\mu}{kT}$$

Bose-Einstein Condensation (BEC)

切入点 $n(\frac{\beta h^2}{2\pi m})^{\frac{3}{2}} = n(\frac{h^2}{2\pi m kT})^{\frac{3}{2}} \ll 1$ (理想气体), 考虑 n 足够大或 T 足够小的情况.

Bose分布 $n_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1} > 0 \Rightarrow \epsilon_i > \mu$ 取粒子平均 $\epsilon_0 = 0$, 则有 $\mu < 0$

考虑粒子平均 (单原子分子). $g(\epsilon) = \frac{2\pi(2m)^{\frac{3}{2}}}{h^3} V \sqrt{\epsilon}$, $N = \int_0^\infty \frac{g(\epsilon)}{e^{\frac{\epsilon - \mu}{kT}} - 1} d\epsilon$

$$N = \frac{2\pi V(2m)^{\frac{3}{2}}}{h^3} \int_0^\infty \frac{\sqrt{\epsilon}}{e^{\frac{\epsilon - \mu}{kT}} - 1} d\epsilon$$

在 $n = \frac{N}{V}$ 不变的情况下, 利用隐函数定理知, $\frac{\partial \mu}{\partial T} = - \frac{\partial n / \partial T}{\partial n / \partial \mu} = - \frac{\frac{\partial}{\partial T} \int_0^\infty \frac{\sqrt{\epsilon}}{e^{\frac{\epsilon - \mu}{kT}} - 1} d\epsilon}{\frac{\partial}{\partial \mu} \int_0^\infty \frac{\sqrt{\epsilon}}{e^{\frac{\epsilon - \mu}{kT}} - 1} d\epsilon} = - \frac{\int_0^\infty \frac{\sqrt{\epsilon} \cdot e^{\frac{\epsilon - \mu}{kT}} \cdot \frac{\epsilon - \mu}{kT} d\epsilon}{(\int_0^\infty \frac{\sqrt{\epsilon} \cdot e^{\frac{\epsilon - \mu}{kT}} d\epsilon)^2}} < 0$

$\Rightarrow \mu$ 随 T 上升而下降, 下降而上升, 但 μ 有上界 0, 所以当 $T \rightarrow T_c$ 时, $\mu \rightarrow 0$

$$\begin{aligned} N &= \frac{2\pi V(2m)^{\frac{3}{2}}}{h^3} \int_0^\infty \frac{\sqrt{\epsilon} d\epsilon}{e^{\frac{\epsilon - \mu}{kT}} - 1} = \frac{2\pi V(2m k T_c)^{\frac{3}{2}}}{h^3} \int_0^\infty \frac{\sqrt{x} dx}{e^x - 1} = \frac{2\pi V}{h^2} \left(\frac{2m k T_c}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \sqrt{x} e^{-x} dx \\ &= 2\pi V \left(\frac{2m k T_c}{h^2} \right)^{\frac{3}{2}} \sum_{k=0}^\infty \int_0^\infty \sqrt{x} e^{-(k+1)x} dx = 2\pi V \left(\frac{2m k T_c}{h^2} \right)^{\frac{3}{2}} \sum_{k=0}^\infty \frac{1}{(k+1)^{\frac{3}{2}}} \Gamma(\frac{3}{2}) \\ &= V \left(\frac{2\pi m k T_c}{h^2} \right)^{\frac{3}{2}} \sum_{k=1}^\infty \frac{1}{k^{\frac{3}{2}}} \approx 2.612 V \left(\frac{2\pi m k T_c}{h^2} \right)^{\frac{3}{2}} \end{aligned}$$

$$T_c = \left(\frac{N}{2.612 V} \right)^{\frac{2}{3}} \cdot \frac{h^2}{2\pi m k} \leftarrow \text{Bose 气体的凝聚温度. [液氦 } n = 2.18 \times 10^{28} \text{ m}^{-3}, m = 6.64 \times 10^{-27} \text{ kg} \text{ (4He). 得 } T_c \approx 3.13 \text{ K}]$$

当 $T < T_c$ 时, $N = N_0 + N_{\epsilon > 0} \leftarrow \begin{matrix} \uparrow \\ \text{局部凝聚化} \end{matrix} 2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \frac{\sqrt{\epsilon} d\epsilon}{e^{\frac{\epsilon - \mu}{kT}} - 1} = \cancel{2\pi V} V \left(\frac{2\pi m k T}{h^2} \right)^{\frac{3}{2}} \sum_{k=1}^\infty \frac{1}{k^{\frac{3}{2}}} \approx 2.612 V \left(\frac{2\pi m k T}{h^2} \right)^{\frac{3}{2}}$

$$\frac{N_{\epsilon > 0}}{N} = \left(\frac{T}{T_c} \right)^{\frac{3}{2}} \Rightarrow N_0 = N - N_{\epsilon > 0} = N \left[1 - \left(\frac{T}{T_c} \right)^{\frac{3}{2}} \right], T < T_c$$

当 $T < T_c$ 时, $\alpha = -\frac{\mu}{kT} = 0$, $\Phi(\alpha, \beta, y) = \underbrace{-g_0 \ln(1 - e^{-\alpha})}_{\uparrow} + 2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \sqrt{\epsilon} \ln(1 - e^{-\alpha - \beta \epsilon}) d\epsilon$

能量 $= 0$, ~~压强 $= 0$~~

热力学贡献 $= 0$

$$= -2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \frac{\sqrt{\epsilon}}{1 - e^{-\alpha - \beta \epsilon}} d\epsilon$$

$$G = N\mu = \bar{E} + PV - TS$$

在 $T < T_c$ 时, $\mu \rightarrow 0$.

$$S = \frac{1}{T} (\bar{E} + PV)$$

$$= -2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \sqrt{\epsilon} \ln(1 - e^{-\alpha - \beta \epsilon}) d\epsilon$$

$$= +2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \frac{\epsilon^{\frac{3}{2}}}{1 - e^{-\alpha - \beta \epsilon}} d\epsilon$$

$$= 2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \frac{\epsilon^{\frac{3}{2}}}{e^{\alpha + \beta \epsilon} - 1} d\epsilon = 2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \frac{\epsilon^{\frac{3}{2}}}{e^{\beta \epsilon} - 1} d\epsilon$$

$$= 2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^\infty \frac{x^{\frac{3}{2}}}{1 - e^{-x}} \cdot e^{-x} dx$$

$$= \frac{4\pi V}{3} \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \sum_{k=0}^\infty \int_0^\infty x^{\frac{3}{2}} e^{-(k+1)x} dx$$

$$= \frac{4\pi V}{3} \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \Gamma(\frac{5}{2}) \sum_{k=0}^\infty \frac{1}{(k+1)^{\frac{5}{2}}} = \frac{4}{3} \pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \cdot \frac{3}{4} \sqrt{\pi} \sum_{k=0}^\infty \frac{1}{(k+1)^{\frac{5}{2}}}$$

$$= V \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}} \sum_{k=0}^\infty \frac{1}{k^{\frac{5}{2}}} \approx 1.341 V \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}}$$

$$\bar{E} = -\frac{\partial \Phi}{\partial \beta} = 1.341 \times 1.5 V \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}} \beta^{-\frac{5}{2}}, P = 1.341 V \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}} \beta^{-\frac{5}{2}}$$

$$S = 1.341 \times 2.5 V \left(\frac{2\pi m T}{h^2} \right)^{\frac{3}{2}} \cdot \frac{1}{T^{\frac{5}{2}}}, C_V = 1.341 \times 1.5 \times 2.5 V \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}} = 1.341 \times 1.5 \times 2.5 V \left(\frac{2\pi m k T}{h^2} \right)^{\frac{3}{2}} = 1.925 N k \left(\frac{T}{T_c} \right)^{\frac{5}{2}}$$

由恒温壁 (温度为 T) 围成的空腔, 腔壁不停发射和吸收光子, 空腔内充满电磁辐射场
当平衡时, 空腔辐射也是 T 温度。

研究空腔辐射的观点 $\rightarrow$ 辐射场分解成许多单色平面电磁波或光子。

$\rightarrow$ 辐射场看作光子气

$\xrightarrow{\text{Einstein}}$ 把波矢 $\vec{k}$ 和频率 ν 单色平面波与动量 $\vec{p}$ 和能量 ϵ 的光子相对应

光子性质: $m_0 = 0$, $\epsilon = h\nu$, $\vec{p} = \frac{h}{2\pi} \vec{k}$, $\vec{k} = \frac{2\pi}{\lambda} \vec{n}$, $\lambda = \frac{c}{\nu} \Rightarrow \epsilon = cp$, $p = |\vec{p}|$

光子是 Bose 子, $s=1$, 自旋简并度是 $J=2$ 。

当空腔体积 V 足够大, 光子能级几乎连续

$$\Omega(\epsilon) = JV \cdot \frac{4\pi}{3} \left(\frac{\epsilon}{h}\right)^3 \Rightarrow \Omega(p) = JV \cdot \frac{4\pi}{3} p^3 \Rightarrow g(p) dp = JV \cdot \frac{4\pi p^2 dp}{h^3} = \frac{8\pi V p^2}{h^3} dp$$

$$p = \frac{h\nu}{c} \Rightarrow g(\nu) d\nu = \frac{8\pi V}{h^3} \left(\frac{h\nu}{c}\right)^2 \cdot \frac{h}{c} d\nu = \frac{8\pi V \nu^2}{c^3} d\nu$$

$$n(\nu) d\nu = g(\nu) f(\nu) d\nu = \frac{8\pi V \nu^2}{c^3} \cdot \frac{1}{e^{\frac{h\nu}{kT}} - 1} d\nu \quad \left[\begin{array}{l} \text{因为光子不停发射和吸收, 所以 } n \text{ 不定} \\ \text{只有 } \sum_i n_i \epsilon_i = E \text{ 是恒定的条件} \end{array} \right]$$

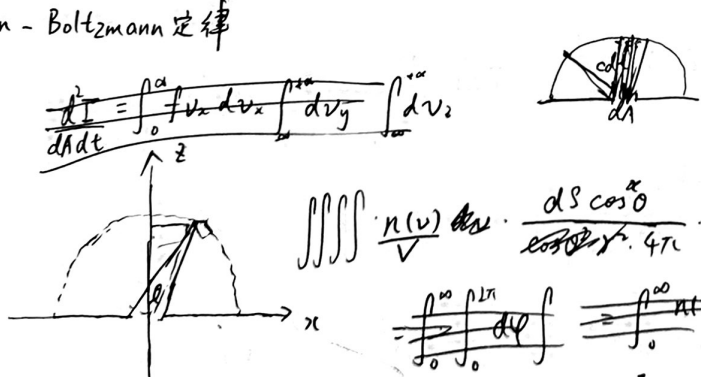
$$\bar{E}(\nu, T) d\nu = n(\nu) h\nu d\nu = \frac{8\pi V}{c^3} \cdot \frac{h\nu^3}{e^{\frac{h\nu}{kT}} - 1} d\nu$$

当低频高温时, $e^{\frac{h\nu}{kT}} - 1 \approx \frac{h\nu}{kT}$, $\bar{E}(\nu, T) d\nu = \frac{8\pi V}{c^3} kT \nu^2 d\nu$ Rayleigh-Jeans 公式

当高频低温时, $\bar{E}(\nu, T) d\nu = \frac{8\pi V}{c^3} \cdot h\nu^3 \cdot e^{-\frac{h\nu}{kT}} d\nu$ Wien 公式

$$\begin{aligned} \bar{E} &= \int_0^\infty \bar{E}(\nu, T) d\nu = \frac{8\pi V}{c^3} \int_0^\infty \frac{h\nu^3}{e^{\frac{h\nu}{kT}} - 1} d\nu = \frac{8\pi V}{(hc)^3} (kT)^4 \int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{8\pi V}{(hc)^3} (kT)^4 \sum_{k=1}^\infty \int_0^\infty x^3 e^{-(k+x)x} dx \\ &= \frac{8\pi V}{(hc)^3} (kT)^4 \cdot \Gamma(4) \cdot \sum_{k=1}^\infty \frac{1}{k^4} = \frac{8\pi V}{(hc)^3} (kT)^4 \cdot 3! \cdot \frac{\pi^4}{90} = \frac{8\pi^5}{15(hc)^3} (kT)^4 V \\ &= \frac{8\pi^5 k^4}{15(hc)^3} VT^4, \quad u = \frac{\bar{E}}{V} = \frac{8\pi^5 k^4}{15(hc)^3} T^4 \end{aligned}$$

Stefan - Boltzmann 定律



$$\begin{aligned} \frac{dI}{dA dt} &= \int_0^\alpha \int_0^{2\pi} \int_0^\infty \int_0^\pi \frac{n(\nu)}{V} \cdot \frac{dS \cos \theta}{r^2} \cdot r^2 \sin \theta \cdot d\varphi \cdot d\theta \cdot dr \cdot d\nu \\ &= \int_0^\infty \int_0^{2\pi} \int_0^\pi \int_0^\alpha \frac{n(\nu)}{V} \sin \theta \cos \theta dr d\theta d\varphi d\nu \\ &= \int_0^\infty \frac{n(\nu)}{V} d\nu \int_0^{2\pi} d\varphi \int_0^\pi \frac{\cos^2 \theta \sin \theta d\theta}{4\pi} \int_0^\alpha dr dS \\ &= \frac{u \cdot 2\pi \cdot c dt dS}{4\pi} \cdot \left[\frac{\sin^3 \theta}{3} \right]_0^\pi = \frac{1}{4} c u dt dS \end{aligned}$$

$$J = \frac{d^2 E_{\text{Emk}}}{dt ds} = \frac{1}{4} cu = \frac{1}{4} \cdot \frac{8\pi^5 k^4}{15(hc)^3} T^4 = \frac{2\pi^5 k^4}{15 h^3 c^2} T^4$$

$$\sigma = 5.670 \times 10^{-8} \text{ J m}^{-2} \text{ s}^{-1} \text{ K}^{-4}$$

$$N = \int_0^\infty n(v) dv = \frac{8\pi V}{c^3} \int_0^\infty \frac{v^2}{e^{\frac{h\nu}{kT}} - 1} dv = \frac{8\pi V (\frac{kT}{hc})^3}{(hc)^3} \int_0^\infty \frac{x^2}{e^x - 1} dx$$

$$= 8\pi V \left(\frac{kT}{hc}\right)^3 \sum_{k=1}^{\infty} \frac{1}{k^3} = 8\pi V \left(\frac{kT}{hc}\right)^3 \cdot 2 \times 1.202 = 8\pi V \left(\frac{kT}{hc}\right)^3 \cdot 2.404$$

$$n = \frac{N}{V} = 2.404 \times 8\pi \left(\frac{k}{hc}\right)^3 T^3$$

Wien 位移律: $n(\lambda) d\lambda = -n(\frac{c}{\lambda}) d(\frac{c}{\lambda}) = n(\frac{c}{\lambda}) \cdot \frac{c}{\lambda^2} d\lambda$

$$n(\lambda) = \frac{8\pi V}{c^3} \frac{(c/\lambda)^2}{e^{\frac{hc}{\lambda kT}} - 1} \cdot \frac{c}{\lambda^2} d\lambda$$

$$\bar{E}(\lambda, T) d\lambda = n(\lambda) \cdot \frac{hc}{\lambda} d\lambda = \frac{8\pi V hc}{\lambda^5} \frac{d\lambda}{e^{\frac{hc}{\lambda kT}} - 1}$$

取极大值 $\frac{\partial \bar{E}(\lambda, T)}{\partial \lambda} \Big|_{\lambda=\lambda_m} = 0$

$$\Leftrightarrow \frac{\partial \ln \bar{E}(\lambda, T)}{\partial \lambda} \Big|_{\lambda=\lambda_m} = 0$$

$$\Rightarrow \frac{\partial}{\partial \lambda} \left(\ln \lambda^5 + \ln \left(e^{\frac{hc}{\lambda kT}} - 1 \right) \right) \Big|_{\lambda=\lambda_m} = 0$$

$$\Rightarrow \frac{5}{\lambda_m} + \frac{e^{\frac{hc}{\lambda_m kT}} \cdot \left(-\frac{1}{\lambda_m^2} \right) \frac{hc}{kT}}{e^{\frac{hc}{\lambda_m kT}} - 1} = 0 \Leftrightarrow 5 - \frac{e^{\frac{hc}{\lambda_m kT}} \cdot \frac{hc}{\lambda_m kT}}{e^{\frac{hc}{\lambda_m kT}} - 1} = 0$$

令 $x = \frac{hc}{\lambda_m kT}$, $5(1 - e^{-x}) = x$, 用图解法可得 $x \approx 4.96$

$$\Rightarrow \lambda_m T = \frac{hc}{x k} \approx \frac{hc}{4.96 k} \approx 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$$

特性函数

$$\Phi(\beta, V) = - \int_0^\infty g(\epsilon) \ln(1 - e^{-\beta \epsilon}) d\epsilon = \int_0^\infty \frac{8\pi \epsilon^2}{(hc)^3} V \ln(1 - e^{-\beta \epsilon}) d\epsilon$$

$$= - \frac{8\pi V}{3(hc)^3} \ln(1 - e^{-\beta \epsilon}) \Big|_0^\infty + \int_0^\infty \frac{8\pi V}{3(hc)^3} \epsilon^3 \cdot \frac{\beta e^{-\beta \epsilon}}{1 - e^{-\beta \epsilon}} d\epsilon$$

$$= \frac{8\pi V}{3(hc)^3} \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx = \frac{8\pi V}{3(hc)^3} \Gamma(4) \sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{8\pi V}{3(hc)^3} \cdot 3! \cdot \frac{\pi^4}{90} = \frac{8\pi^5 V}{45(hc)^3}$$

$$\bar{E} = - \frac{\partial \Phi}{\partial \beta} = \frac{8\pi^5 V}{15(hc)^3} \beta^{-4} = \frac{8\pi^5 k^4}{15(hc)^3} VT^4 = bVT^4$$

$$\bar{P} = \frac{1}{\beta} \frac{\partial \Phi}{\partial V} = \frac{8\pi^5 k^4}{45(hc)^3} T^4 = \frac{1}{3} \bar{E} = \frac{1}{3} bVT^4$$

$$S = k \left(\Phi - \beta \frac{\partial \Phi}{\partial \beta} - \alpha \frac{\partial \Phi}{\partial \alpha} \right) = k \left(\frac{8\pi^5 V}{45(hc)^3} \beta^{-3} + \frac{8\pi^5 V}{45(hc)^3} \beta^{-3} \right) = \frac{4}{3} \cdot \frac{8\pi^5 k^4}{15(hc)^3} T^3 = \frac{4}{3} bVT^3$$

$$\bar{Q}_V = \frac{\partial \bar{E}}{\partial T} = 4 \cdot \frac{8\pi^5 k^4}{15(hc)^3} VT^3 = 4bVT^3$$

$$Q = \bar{E} + P\bar{V} - T\bar{S} = bVT^4 + \frac{1}{3}bVT^4 - \frac{4}{3}bVT^4 = 0 \Rightarrow \mu = \frac{G}{N} = 0$$

$$F = \bar{E} - T\bar{S} = -\frac{1}{3}bVT^4 = -\frac{1}{3}\bar{E}$$

聲子氣體:

晶格振動熱容量 $\rightarrow$ Einstein: $C_v = 3Nk_B \left(\frac{\theta_E}{T}\right)^2 e^{-\frac{\theta_E}{T}}$
(低溫) $\rightarrow$ 實驗: $C_v \propto T^3$

修正 Einstein 理論, Debye 理論 $\rightarrow$ (1) 簡正振動波長 $\gg$ 晶格常數時, 可忽略晶格的 discreteness, 把固體看成均勻同性的連續介質, 原子振動形成的各個簡正振動頻率, 的彈性縱波

(1) 彈性縱波頻率在 0 到 ν_D (截止頻率) 之間, N 個原子共 $3N$ 個自由度, 等價於 $3N$ 個簡正頻率的彈性縱波.

晶格振動 $\rightarrow$ 宏觀表現是聲波

$\rightarrow$ 在低溫下, 被激發的主要是低頻聲波 $\rightarrow$ 波長 $\gg$ 晶格常數, 採用 Debye 理論

聲子: 具有一定頻率和波長的聲波與具有一定能量與動量的粒子相當

假設聲波波速是 v , 頻率是 ν , 波長是 λ , 則

$$E = h\nu, \quad \vec{p} = \frac{h}{\lambda} \vec{k}, \quad \vec{k} = \frac{2\pi}{\lambda} \vec{n}, \quad \lambda = \frac{v}{\nu} \Rightarrow \vec{p} = \frac{h\nu}{v} \vec{n}$$

$$E = \sum_{i=1}^{3N} (n_i + \frac{1}{2}) h\nu_i + \phi_0 \leftarrow \text{平衡相互作用能}$$

$$= \sum_{i=1}^{3N} n_i h\nu_i + \sum_{i=1}^{3N} \frac{1}{2} h\nu_i + \phi_0$$

$\leftarrow$ 零點能

$$= \sum_{i=1}^{3N} n_i h\nu_i + E_0, \quad \phi_0 < 0, \quad |\phi_0| > \sum_{i=1}^{3N} \frac{1}{2} h\nu_i$$

$\leftarrow$ 固體結合能

$\leftarrow$ 激發能

對比諧振子能量 $\epsilon_n = (n + \frac{1}{2}) h\nu$
相當於在 ν 頻率上出現了 n 個聲子
因為這些諧振子是獨立地振動,
所以聲子是近獨立粒子.

聲子和光子 $\rightarrow$ 共同: (1) 不斷產生和消滅, 所以 $\alpha = 0$
(2) Bose 子

$\rightarrow$ 不同: 光子頻率可達無限, 聲子不能

聲子: $\rightarrow$ 橫波: $J=2$ (兩個獨立振動方向 y 和 z), 波速為 v_t , $g(p)dp = \frac{8\pi V}{h^3} p^2 dp$

$\rightarrow$ 縱波: $J=1$ (振動方向只有 x), 波速為 v_l , $g(p)dp = \frac{4\pi V}{h^3} p^2 dp$

$$g(\nu)d\nu = 4\pi V \left(\frac{2}{v_t^3} + \frac{1}{v_l^3} \right) \nu^2 d\nu = B \nu^2 d\nu$$

$$\text{修正 } g(\nu)d\nu = \begin{cases} B \nu^2 d\nu & 0 \leq \nu \leq \nu_D \\ 0 & \nu > \nu_D \end{cases}, \quad \text{振動模式總數} = 3N = \int_0^{\nu_D} g(\nu)d\nu = \frac{B}{3} \nu_D^3 \Rightarrow \nu_D = \left(\frac{9N}{B} \right)^{\frac{1}{3}}$$

$$\Rightarrow \begin{cases} \frac{9N}{\nu_D^3} \nu^2 d\nu & 0 \leq \nu \leq \nu_D \\ 0 & \nu > \nu_D \end{cases}$$

$$\text{Debye 假設的合理性: 不妨設 } v_t = v_l = v, \quad \nu_D = \left(\frac{3N}{4\pi V} \right)^{\frac{1}{3}} v, \quad \lambda_{\min} = \lambda_D = \frac{v}{\nu_D} = \left(\frac{4\pi V}{3N} \right)^{\frac{1}{3}} \approx 1.61 \frac{a}{\left(\frac{N}{V} \right)^{\frac{1}{3}}}$$

$$n(\nu)d\nu = \frac{9N}{\nu_D^3} \cdot \frac{1}{e^{\frac{h\nu}{kT}} - 1} \nu^2 d\nu, \quad E(T, V) = E_0 + \int_0^{\nu_D} \frac{9N}{\nu_D^3} \cdot \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1} d\nu$$

$$= E_0 + \frac{9N}{(h\nu_D)^3} (kT)^4 \frac{\pi^4}{15} = E_0 + \frac{9N}{(h\nu_D)^3} (kT)^4 \frac{\pi^4}{15}$$

$$= E_0 + \frac{9N}{(h\nu_D)^3} (kT)^4 \int_0^{\frac{h\nu_D}{kT}} \frac{x^3}{e^x - 1} dx$$

$$= E_0 + \frac{9N}{(h\nu_D)^3} (kT)^4 \frac{\pi^4}{15}$$

$$\bar{E}(T, V) = E_0 + \frac{9N}{V_0^3} \int_0^{V_0} \frac{h\nu^3}{e^{\beta h\nu} - 1} d\nu \quad C_V = \frac{\partial \bar{E}}{\partial T} = \frac{9N}{V_0^3} \int_0^{V_0} \frac{e^{\frac{h\nu}{kT}} \cdot \frac{h\nu}{kT} \cdot h\nu^3}{(e^{\frac{h\nu}{kT}} - 1)^2} d\nu$$

$$= \frac{9N}{V_0^3} \int_0^{V_0} \frac{k e^{\frac{h\nu}{kT}} \left(\frac{h\nu}{kT}\right)^2}{(e^{\frac{h\nu}{kT}} - 1)^2} \nu^2 d\nu = \frac{9N}{V_0^3} \int_0^{V_0} k \underbrace{E\left(\frac{h\nu}{kT}\right)}_{\substack{\uparrow \\ \text{Einstein函数}}} \nu^2 d\nu$$

記特征溫度 $\theta_0 = \frac{h\nu_0}{k}$, $y = \frac{\theta_0}{T} = \frac{h\nu_0}{kT}$

~~$C_V(\theta_0)$~~ $C_V(y) = \frac{9Nk}{V_0^3} \left(\frac{kT}{h}\right)^3 \int_0^y \frac{x^3 e^x}{(e^x - 1)^2} x^2 dx = \frac{9Nk}{y^3} \int_0^y \frac{x^4 e^x}{(e^x - 1)^2} dx$

$$= \frac{9Nk}{y^3} \int_0^y x^4 d\left[-\frac{1}{e^x - 1}\right] = \frac{9Nk}{y^3} \left[-\frac{y^4}{e^y - 1} + \int_0^y \frac{4x^3}{e^x - 1} dx\right]$$

$$= 3Nk \left[\underbrace{\frac{3}{y^3} \int_0^y \frac{x^3 dx}{e^x - 1}}_{\substack{\uparrow \\ \text{Debye 函数}}} \cdot 4 - \frac{3y}{e^y - 1} \right] = 3Nk \left[4D(y) - \frac{3y}{e^y - 1} \right]$$

(1) 高溫極限 $T \gg \theta_0$, $y \rightarrow 0$.

$$C_V(y) \approx 3Nk \left[\frac{3}{y^3} \int_0^y x^3 dx - \frac{3y}{y} \right] = 3Nk$$

(2) 低溫極限 $T \ll \theta_0$, ~~$y \rightarrow \infty$~~ $y \gg 1$

$$C_V(y) = \frac{9Nk}{y^3} \left[\int_0^\infty \frac{x^4 e^x}{(e^x - 1)^2} dx - \int_y^\infty \frac{x^4 e^x}{(e^x - 1)^2} dx \right]$$

$$= \frac{9Nk}{y^3} \left[\frac{4\pi^4}{15} - \sum_{k=1}^\infty \int_y^\infty x^4 e^{-kx} dx \right]$$

$$\approx \frac{9Nk}{y^3} \left[\frac{4\pi^4}{15} - \int_y^\infty x^4 e^{-x} dx \right]$$

Debye 定律 $\rightarrow \approx \frac{9Nk}{y^3} \left[\frac{4\pi^4}{15} - y^4 e^{-y} \right] = 3Nk \left(\frac{4\pi^4}{15y^3} - y e^{-y} \right) \approx 3Nk \cdot \frac{4\pi^4}{15y^3} \propto T^3$

修正 Debye 理論, Born 理論 $\rightarrow$ 以分子為整體振動, 振頻低 (聲頻振動)

$\rightarrow$ 以分子內各原子相對振動, 振頻高 (光頻振動)

物性函数 $\Phi(\beta, V) = - \int_0^{V_0} B \nu^2 \ln(1 - e^{-\beta h\nu}) d\nu$

~~聲子氣壓 $\frac{1}{\beta} \frac{\partial \Phi}{\partial V} = - \frac{1}{\beta} \frac{\partial B}{\partial V} \int_0^{V_0} \frac{\nu^2}{e^{\beta h\nu} - 1} d\nu = - \frac{1}{\beta} 4\pi \left(\frac{1}{V_0^3} + \frac{1}{V_0^3} \right) \int_0^{V_0} \nu^2 \ln(1 - e^{-\beta h\nu}) d\nu$~~

$$\Phi(\beta, V) = - \int_0^{V_0} B \nu^2 \ln(1 - e^{-\beta h\nu}) d\nu = - \frac{9N}{V_0^3} \int_0^{V_0} \nu^3 \ln(1 - e^{-\beta h\nu}) d\nu$$

~~$P = - \frac{1}{\beta} \frac{\partial \Phi}{\partial V} = - \frac{1}{\beta} \cdot 4\pi \left(\frac{2}{V_0^3} + \frac{1}{V_0^3} \right) \int_0^{V_0} \nu^2 \ln(1 - e^{-\beta h\nu}) d\nu$~~

$$P = \frac{1}{\beta} \frac{\partial \Phi}{\partial V} = \frac{1}{\beta} \frac{\partial \Phi}{\partial V_0} \cdot \frac{\partial V_0}{\partial V} = \frac{9N}{\beta} \left(-3 \cdot \frac{1}{V_0^3} \int_0^{V_0} \nu^3 \ln(1 - e^{-\beta h\nu}) d\nu + \frac{1}{V_0^3} \ln(1 - e^{-\beta h\nu_0}) \right) \cdot \left(-\frac{1}{3} \frac{V_0}{V} \right)$$

$$= - \frac{9NkT}{V_0^3} \int_0^{V_0} \nu^3 \ln(1 - e^{-\beta h\nu}) d\nu + \frac{3NkT}{V} \ln(1 - e^{-\beta h\nu_0})$$

$$= - \frac{3NkT}{V_0^3} \left(V_0^3 \ln(1 - e^{-\beta h\nu_0}) - \int_0^{V_0} \frac{\beta h e^{-\beta h\nu}}{1 - e^{-\beta h\nu}} \nu^3 d\nu \right) + \frac{3NkT}{V} \ln(1 - e^{-\beta h\nu_0})$$

$$= \frac{3N}{V_0^3} \int_0^{V_0} \frac{h\nu^3}{e^{\beta h\nu} - 1} d\nu = \frac{1}{3V} (\bar{E}(T, V) - E_0)$$

$$P_0(T, V) = - \left(\frac{\partial F(T, V)}{\partial V} \right)_T = - \frac{\partial \bar{E}_0(V)}{\partial V}, \quad \bar{E}_0 = \phi_0 + \frac{1}{2} \int_0^{V_0} h \nu g(\nu) d\nu$$

Fermi 氣能

常遇情况 $\rightarrow n \left(\frac{h^2}{2\pi m kT} \right) \gg 1$, 完全简并 ($T=0$)

$\rightarrow n_i = \frac{g_i}{e^{(\epsilon_i - \mu)/kT} + 1}$, $f_i = \frac{n_i}{g_i} = \frac{1}{e^{(\epsilon_i - \mu)/kT} + 1}$, $\alpha + \beta \epsilon_i = \frac{\epsilon_i - \mu}{kT}$

当 $T=0K$ 时, $f_i = \begin{cases} 1, & \epsilon_i < \mu \\ 0, & \epsilon_i > \mu \end{cases}$, 定义 $\mu = \epsilon_F$ 称为 ~~费米~~ Fermi 能量

$g(\epsilon) d\epsilon = 2\pi V J \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \sqrt{\epsilon} d\epsilon = CV \sqrt{\epsilon} d\epsilon$

$N = \int_0^\infty n(\epsilon) d\epsilon = \int_0^\infty f(\epsilon) g(\epsilon) d\epsilon = \int_0^{\epsilon_F} \frac{2\pi V J \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \sqrt{\epsilon}}{e^{\frac{\epsilon - \mu}{kT} + 1}} d\epsilon$

$= \int_0^{\epsilon_F} 2\pi V J \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \sqrt{\epsilon} d\epsilon = \frac{2}{3} \cdot 2\pi V J \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \epsilon_F^{\frac{3}{2}} = \frac{2}{3} CV \epsilon_F^{\frac{3}{2}}$

$\epsilon_F = \left(\frac{3N}{2CV} \right)^{\frac{2}{3}} = \left(\frac{3N}{4\pi V J} \right)^{\frac{2}{3}} \cdot \frac{h^2}{2m}$, 对于 $S = \frac{1}{2}$, $J = 2S + 1 = 2$ 的电子系统

$\epsilon_F = \left(\frac{3N}{8\pi V} \right)^{\frac{2}{3}} \cdot \frac{h^2}{2m}$

$\bar{E}_0 = \int_0^{\epsilon_F} \epsilon f(\epsilon) g(\epsilon) d\epsilon = \int_0^{\epsilon_F} CV \epsilon^{\frac{3}{2}} d\epsilon = \frac{2}{5} CV \epsilon_F^{\frac{5}{2}} = \frac{3}{5} N \epsilon_F = AV^{\frac{2}{3}}$

$\bar{E}_0 = \frac{\bar{E}_0}{N} = \frac{3}{5} \epsilon_F$, $\bar{F}_0 = \bar{E}_0 - T_0 S_0 = \bar{E}_0 = \frac{3}{5} N \epsilon_F$

~~$P_0 = \frac{\partial \bar{F}_0}{\partial V} = - \frac{\partial \bar{F}_0}{\partial V} = - \frac{2}{5} \frac{CV \epsilon_F^{\frac{5}{2}}}{V} = - \frac{2}{5} \frac{N \epsilon_F}{V}$~~ $P = \frac{2}{3} \frac{E_0}{V} = \frac{2}{5} \frac{N \epsilon_F}{V}$ (习题 7.1)

$= - \frac{\partial \bar{F}_0}{\partial V} = - \frac{2}{3} \frac{AV^{-\frac{2}{3}}}{V} = \frac{2}{3} \frac{E_0}{V} = \frac{4}{15} C \epsilon_F^{\frac{5}{2}}$

记特征温度 $T_F = \frac{\epsilon_F}{k}$; 以金属为例, 电子数密度 $n = 4 \times 10^{28} m^{-3}$, $m = 9.1 \times 10^{-31} kg$

$\epsilon_F \approx 6.9 \times 10^{-19} J$, $\bar{E}_0 \approx 4.2 \times 10^{-19} J$, $\bar{P}_0 \sim 10^{10} Pa \sim 10^5 atm$

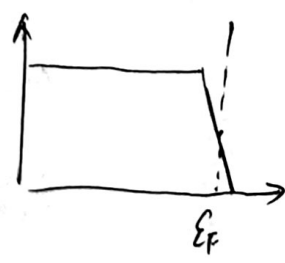
在 $T=0K$ 时, $W_F = 1 \Rightarrow S = 0$

热力学第三定律, $\lim_{T \rightarrow 0} C_v = 0$

Bose: $\bar{E}, \bar{P}, P, S = 0$ ($T=0K$)

Fermi: $\bar{E}, \bar{P}, P, \uparrow, S = 0$

有限温度分析, $T > 0$ 但 $T \ll T_F$, $f(\epsilon) = \frac{1}{e^{\frac{\epsilon - \mu}{kT} + 1}} \approx \begin{cases} 1, & \epsilon < \mu - kT \\ 0 \sim 1, & -kT < \epsilon - \mu < kT \\ 0, & \epsilon > \mu + kT \end{cases}$



~~有效电子数 $N_{eff} = \int_{\epsilon_F - kT}^{\epsilon_F + kT} CV \sqrt{\epsilon} \cdot \left(\frac{\epsilon - \epsilon_F}{kT} \right)^{\frac{1}{2}} d\epsilon$~~

$N_{有效} = \frac{2kT}{\epsilon_F} N$, $\Delta E = \bar{E}(T, V) - \bar{E}(0, V) \approx N_{有效} kT = \left(N \cdot \frac{2kT}{\epsilon_F} \right) \cdot kT \propto T^2$

热运动的平均能量 $\propto kT$

热容量 $C_v = 4NR \frac{kT}{\epsilon_F} \propto T$

$\propto \frac{3}{2} NR$

$$Q_e = \int_0^\infty f(\epsilon) \epsilon^l d\epsilon$$

$$= \frac{1}{l+1} \int_0^\infty \epsilon^{l+1} \left(-\frac{df}{d\epsilon}\right) d\epsilon \quad -\frac{df}{d\epsilon} = \frac{\frac{1}{\epsilon T} e^{\frac{\epsilon}{kT}}}{\left(e^{\frac{\epsilon}{kT}} + 1\right)^2}$$

$$= \frac{1}{l+1} \int_0^\infty \sum_{n=0}^{\infty} \left(\frac{\mu}{kT}\right)^n \mu^{l-n+1} \epsilon^n \cdot \frac{\frac{1}{\epsilon T} e^{\frac{\epsilon}{kT}}}{\left(e^{\frac{\epsilon}{kT}} + 1\right)^2} d\epsilon \quad \text{因为 } \frac{\mu}{kT} \gg 1$$

$$\approx \frac{1}{l+1} \int_{-\infty}^{+\infty} \sum_{n=0}^{\infty} \left(\frac{\mu}{kT}\right)^n \mu^{l-n+1} x^n \cdot \frac{\frac{1}{x T} e^x}{(e^x + 1)^2} dx \quad (kT)^{n+1} dx$$

$$= \frac{1}{l+1} \sum_{n=0}^{\infty} \int_{-\infty}^{+\infty} \left(\frac{\mu}{kT}\right)^{l-n+1} (kT)^n x^n \cdot \frac{e^x}{(e^x + 1)^2} dx = \frac{1}{l+1} \mu^{l+1} \int_{-\infty}^{+\infty} \left(\frac{\mu}{kT}\right)^n \frac{x^n e^x}{(e^x + 1)^2} dx$$

$$= \frac{1}{l+1} \mu^{l+1} \left[1 + \frac{1}{6} l(l+1) \left(\frac{\pi kT}{\mu}\right)^2\right]$$

$$N = \int_0^\infty f(\epsilon) g(\epsilon) d\epsilon = CV \int_0^\infty f(\epsilon) \epsilon^{\frac{1}{2}} d\epsilon = CV Q_{\frac{1}{2}} = \frac{2}{3} \mu^{\frac{3}{2}} \left[1 + \frac{1}{8} \left(\frac{\pi kT}{\mu}\right)^2 + \dots\right] CV = \frac{2}{3} CV \epsilon_F^{\frac{3}{2}}$$

$$\mu = \left[1 + \frac{1}{8} \left(\frac{\pi kT}{\mu}\right)^2 + \dots\right]^{-\frac{2}{3}} \epsilon_F \approx \left[1 + \frac{1}{8} \left(\frac{\pi kT}{\epsilon_F}\right)^2 + \dots\right]^{-\frac{2}{3}} \epsilon_F \approx \left[1 - \frac{1}{12} \left(\frac{\pi kT}{\epsilon_F}\right)^2\right] \epsilon_F$$

Fermi 氣體的化學勢 μ 可正可負。 $\begin{cases} T=0, \mu > 0 \\ T \uparrow, \mu \downarrow \\ T \gg T_F, \text{氣體服從經典分布, } \mu < 0 \end{cases}$

$$\bar{E} = \int_0^\infty f(\epsilon) g(\epsilon) \epsilon d\epsilon = \int_0^\infty f(\epsilon) CV \epsilon^{\frac{3}{2}} d\epsilon = CV Q_{\frac{3}{2}} = \frac{2}{5} \mu^{\frac{5}{2}} \left[1 + \frac{5}{8} \left(\frac{\pi kT}{\mu}\right)^2 + \dots\right] = \frac{2}{5} \epsilon_F^{\frac{5}{2}}$$

$$\approx \frac{2}{5} CV \epsilon_F^{\frac{5}{2}} \left[1 - \frac{5}{24} \left(\frac{\pi kT}{\epsilon_F}\right)^2 + \dots\right] \left[1 + \frac{5}{8} \left(\frac{\pi kT}{\epsilon_F}\right)^2 + \dots\right]$$

$$\approx \frac{2}{5} CV \epsilon_F^{\frac{5}{2}} \left[1 - \frac{5}{24} \left(\frac{\pi kT}{\epsilon_F}\right)^2 + \dots\right] \left[1 + \frac{5}{8} \left(\frac{\pi kT}{\epsilon_F}\right)^2 + \dots\right]$$

$$= \bar{E}_0 \left[1 - \frac{5}{24} \left(\frac{\pi kT}{\epsilon_F}\right)^2\right] \left[1 + \frac{5}{8} \left(\frac{\pi kT}{\epsilon_F}\right)^2\right]$$

$$\approx \bar{E}_0 \left[1 + \frac{5}{12} \left(\frac{\pi kT}{\epsilon_F}\right)^2 + \dots\right]$$

$$C_V = \frac{\partial E}{\partial T} = \frac{2}{5} CV \epsilon_F^{\frac{5}{2}} \cdot \frac{5}{12} \cdot \left(\frac{\pi kT}{\epsilon_F}\right) \cdot 2 \left(\frac{\pi kT}{\epsilon_F}\right)$$

$$= \frac{2}{5} N \epsilon_F \cdot \frac{5}{6} \left(\frac{\pi kT}{\epsilon_F}\right)^2$$

$$= \frac{2}{5} N \epsilon_F \cdot \frac{\pi^2}{2} \cdot \left(\frac{kT}{\epsilon_F}\right) \approx 4.9 N k \left(\frac{kT}{\epsilon_F}\right)$$

特性函數 $\Phi(\alpha, \beta, V) = \int_0^\infty f(\epsilon) \ln(1 - e^{-\alpha - \beta \epsilon}) d\epsilon$

$$= CV \int_0^\infty \ln(1 + e^{-\alpha - \beta \epsilon}) \sqrt{\epsilon} d\epsilon$$

$$= \frac{2}{3} CV \int_0^\infty \frac{e^{-\alpha - \beta \epsilon} \cdot \beta}{1 + e^{-\alpha - \beta \epsilon}} \epsilon^{\frac{1}{2}} d\epsilon = \frac{2}{3} CV \beta \int_0^\infty f(\epsilon) \epsilon^{\frac{1}{2}} d\epsilon$$

$$= \frac{2}{3} CV \beta Q_{\frac{1}{2}} = \frac{2}{3} CV \beta \cdot \frac{2}{5} \mu^{\frac{5}{2}} \left[1 + \frac{5}{8} \left(\frac{\pi kT}{\mu}\right)^2\right]$$

$$= \frac{4}{15} CV \beta \left(-\frac{\alpha}{\beta}\right)^{\frac{1}{2}} \left[1 + \frac{5}{8} \left(\frac{\pi}{\alpha}\right)^2\right]$$

$$P = \frac{1}{\beta} \frac{\partial \Phi}{\partial V} = \frac{4}{15} C \left(-\frac{\alpha}{\beta}\right)^{\frac{1}{2}} \left[1 + \frac{5}{8} \left(\frac{\pi}{\alpha}\right)^2\right] = \frac{4}{15} C \mu^{\frac{5}{2}} \left[1 + \frac{5}{8} \left(\frac{\pi kT}{\mu}\right)^2\right] = \bar{P}_0 \left(1 - \frac{5}{24} \left(\frac{\pi kT}{\epsilon_F}\right)^2\right) \left(1 + \frac{5}{8} \left(\frac{\pi kT}{\epsilon_F}\right)^2\right) \approx \bar{P}_0 \left(1 + \frac{5}{12} \left(\frac{\pi kT}{\epsilon_F}\right)^2\right)$$

$$\text{熵 } S = k \left(\Phi - \mu \frac{\partial \Phi}{\partial \mu} - \alpha \frac{\partial \Phi}{\partial \alpha} \right)$$

$$= k \left(\frac{4}{15} CV \left(-\frac{\alpha}{\mu} \right)^{\frac{1}{2}} \left(1 + \frac{1}{2} \left(\frac{\alpha}{\mu} \right)' \right) + \frac{2}{3} CV \left(-\frac{\alpha}{\mu} \right)^{\frac{1}{2}} \left(1 + \frac{1}{2} \left(\frac{\alpha}{\mu} \right)' \right) + \frac{4}{15} CV \left[\frac{1}{2} \left(-\frac{\alpha}{\mu} \right)^{\frac{1}{2}} \left(1 + \frac{1}{2} \left(-\frac{\alpha}{\mu} \right)' \right) + \left(-\frac{\alpha}{\mu} \right)^{\frac{1}{2}} \left(\frac{1}{2} \cdot \mu \cdot \frac{1}{\mu^2} \right) \right] \right)$$

$$\text{熵 } S = k \left(\Phi - \mu \frac{\partial \Phi}{\partial \mu} - \alpha \frac{\partial \Phi}{\partial \alpha} \right)$$

$$= k \left(\Phi - \mu \frac{\partial \Phi}{\partial \mu} - \alpha \frac{\partial \Phi}{\partial \alpha} \right) = k \left[\frac{4}{15} CV \frac{1}{\mu^{\frac{1}{2}}} \left(1 + \frac{1}{2} \left(\frac{\alpha}{\mu} \right)' \right) + \frac{1}{\mu^{\frac{1}{2}}} \left[\frac{2}{3} CV \left(1 + \frac{1}{2} \left(\frac{\alpha}{\mu} \right)' \right) - \frac{1}{2} + CV \left(1 - \frac{1}{2} \left(\frac{\alpha}{\mu} \right)' \right) \right] \right]$$

$$= k \frac{1}{\mu^{\frac{1}{2}}} CV \frac{1}{\mu^{\frac{1}{2}}} \left(\frac{\alpha}{\mu} \right)' = \frac{2}{3} CV \alpha^{\frac{1}{2}} \mu^{-\frac{3}{2}}$$

$$= k \frac{2}{3} CV \epsilon_r^{\frac{1}{2}} \cdot \frac{1}{2 \epsilon_r} \cdot T = N k \left(\frac{\pi^2 V T}{3 \epsilon_r} \right) \propto T$$

$$8.1 \quad S = k \left(\Phi - \beta \frac{\partial \Phi}{\partial \beta} - \alpha \frac{\partial \Phi}{\partial \alpha} \right)$$

$$W_A \{n_i\} = \prod_i \binom{n_i + g_i - 1}{g_i - 1}, \quad W_F \{n_i\} = \prod_i \binom{g_i}{n_i}$$

$$\ln W_F \{n_i\} = \sum_i \left(n_i \ln \left(\frac{g_i}{n_i} + 1 \right) + g_i \ln \left(1 + \frac{n_i}{g_i} \right) \right), \quad \text{代入 } n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i} \pm 1}$$

$$= \sum_i \left(n_i (\alpha + \beta \epsilon_i) \pm g_i \ln \left(\frac{e^{\alpha + \beta \epsilon_i} \pm 1}{e^{\alpha + \beta \epsilon_i}} \right) \right)$$

$$= \sum_i \left(n_i \alpha + \beta n_i \epsilon_i \pm g_i \ln (1 \pm e^{-\alpha - \beta \epsilon_i}) \right)$$

$$= \alpha N + \beta \bar{E} + \Phi$$

$$= \Phi - \beta \frac{\partial \Phi}{\partial \beta} - \alpha \frac{\partial \Phi}{\partial \alpha}$$

$$\Rightarrow S = k \left(\Phi - \beta \frac{\partial \Phi}{\partial \beta} - \alpha \frac{\partial \Phi}{\partial \alpha} \right) = k \ln W_F \{n_i\}$$

8.4 = 维氣體 $\Omega(\epsilon) = JS\pi \cdot 2m\epsilon$, $g(\epsilon) d\epsilon = \frac{JS 2m\pi}{h^2} d\epsilon$

若二維氣體有 BEC 現象, 則

$$n = \frac{N}{S} = \int_0^\infty \frac{g(\epsilon) d\epsilon}{e^{\frac{\epsilon}{kT}} - 1} \quad \text{收敛}$$

$$= \int_0^\infty \frac{\frac{2m\pi JS}{h^2} d\epsilon}{e^{\frac{\epsilon}{kT}} - 1} = \frac{2m\pi JS}{h^2} \int_0^\infty \frac{e^{-\frac{\epsilon}{kT}} d\epsilon}{1 - e^{-\frac{\epsilon}{kT}}} = \frac{2m\pi kT_c JS}{h^2} \int_0^\infty \frac{e^{-x} dx}{1 - e^{-x}}$$

$$= - \frac{2m\pi kT_c JS}{h^2} \int_0^\infty \frac{1}{1 - e^{-x}} d(e^{-x})$$

$$= \frac{2m\pi kT_c JS}{h^2} \ln(1 - e^{-x}) \Big|_0^\infty = +\infty$$

所以化學勢 μ 不可能達 0^- , 不存在 BEC

$$8.10 \quad \bar{E} = \frac{8\pi^5 k^4}{15(hc)^3} VT^4, \quad C_V = \frac{\partial \bar{E}}{\partial T} = \frac{4}{3} b VT^3, \quad S = \int \frac{C_V}{T} dT = 4bV \int T^2 dT = \frac{4}{3} b VT^3 = \frac{32\pi^5 k^4}{45(hc)^3} VT^3 = \frac{4\pi^5 k^4}{45(hc)^3} VT^3$$

$$8.13 \quad \epsilon_F = \left(\frac{3n}{8\pi} \right)^{\frac{2}{3}} \cdot \frac{h^2}{2m} = 8.8631 \times 10^{-19} \text{ J} = 5.5395 \text{ eV}$$

$$v_F = \sqrt{\frac{2\epsilon_F}{m}} = 1.396 \times 10^6 \text{ m/s}, \quad p = \frac{2}{5} n \epsilon_F = 2.092 \times 10^{10} \text{ Pa}$$

8.18 極端相對論 $\epsilon = cp$

$$\Omega(\epsilon) = JV \cdot \frac{4}{3} \pi \left(\frac{\epsilon}{c} \right)^3, \quad g(\epsilon) d\epsilon = \frac{JV}{h^3} \cdot 4\pi \left(\frac{\epsilon}{c} \right)^2 \cdot \frac{1}{c} d\epsilon = \frac{8\pi V}{(hc)^3} \epsilon^2 d\epsilon$$

$$n = \frac{N}{V} = \int_0^{\epsilon_F} \frac{8\pi}{(hc)^3} \epsilon^2 d\epsilon = \frac{8\pi}{3(hc)^3} \epsilon_F^3 \Rightarrow \epsilon_F = \left(\frac{3n}{8\pi} \right)^{\frac{1}{3}} hc$$

$$\bar{E} = \int_0^{\epsilon_F} \epsilon g(\epsilon) d\epsilon = \int_0^{\epsilon_F} \frac{8\pi V}{(hc)^3} \epsilon^3 d\epsilon = \frac{2\pi V}{(hc)^3} \epsilon_F^4 = \frac{2\pi V}{(hc)^3} \cdot \left(\frac{3n}{8\pi} \right) (hc)^3 \epsilon_F = \frac{3}{4} N \epsilon_F$$

$$p = \frac{2}{3} \frac{\bar{E}}{V} = \frac{1}{2} n \epsilon_F \quad [\text{這個是題 7.1 的公式}]$$

補 8.19

(1)

$$\Omega(\epsilon) = J \cdot S \cdot \pi \cdot 2m\epsilon \Rightarrow g(\epsilon) d\epsilon = \frac{2m\pi J S}{h^2} d\epsilon$$

$$n = \frac{N}{S} = \int_0^{\epsilon_F} \frac{g(\epsilon)}{S} d\epsilon = \int_0^{\epsilon_F} \frac{2m\pi J}{h^2} d\epsilon = \frac{2m\pi J}{h^2} \epsilon_F \Rightarrow \epsilon_F = \frac{h^2}{2m\pi J} n = \frac{h^2}{4\pi m} n$$

$$E = \int_0^{\epsilon_F} \epsilon g(\epsilon) d\epsilon = \int_0^{\epsilon_F} \frac{2m\pi J S}{h^2} \epsilon d\epsilon = \frac{Jm\pi S}{h^2} \epsilon_F^2 = \frac{2\pi m S}{h^2} \epsilon_F^2 = \frac{2\pi m S}{h^2} \left(\frac{h^2}{4\pi m} n \right)^2 = \frac{1}{2} n S \epsilon_F = \frac{1}{2} N \epsilon_F$$

$$P = \frac{\partial E}{\partial S} = \frac{1}{2} n \epsilon_F$$

(2) 補充題 1.

$$(1) \Omega(\epsilon) = V \cdot \frac{4}{3} \pi (2m\epsilon)^{\frac{3}{2}} \Rightarrow g(\epsilon) d\epsilon = \frac{2\pi V}{h^3} (2m)^{\frac{3}{2}} \sqrt{\epsilon} d\epsilon$$

首先, Bose 分布滿足 $n_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1} > 0 \Rightarrow \epsilon_i > \mu$. 取平均 $\epsilon_0 = 0$. 則 $\mu \leq 0$

再將利用隱函數定理及 $dN = 0$ 的條件, 可得

$$\frac{d\mu}{dT} = - \frac{(\partial N / \partial T)}{(\partial N / \partial \mu)} = - \frac{\int_0^{\infty} \frac{\sqrt{\epsilon} \exp(\frac{\epsilon - \mu}{kT})}{(\exp(\frac{\epsilon - \mu}{kT}) - 1)^2} d\epsilon}{\int_0^{\infty} \frac{\sqrt{\epsilon} \exp(\frac{\epsilon - \mu}{kT})}{(\exp(\frac{\epsilon - \mu}{kT}) - 1)^2} d\epsilon} < 0$$

即溫度下降時, 化學勢 μ 上升, 因溫度 T 下降至 T_c 時, $\mu \rightarrow 0$

$$N = \int_0^{\infty} \frac{g(\epsilon)}{\exp(\frac{\epsilon}{kT_c}) - 1} d\epsilon = 2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^{\infty} \frac{\epsilon^{\frac{1}{2}}}{\exp(\frac{\epsilon}{kT_c}) - 1} d\epsilon = 2\pi V \left(\frac{2mkT_c}{h^2} \right)^{\frac{3}{2}} \int_0^{\infty} \frac{x^{\frac{1}{2}}}{\exp(x) - 1} dx$$

$$= 2\pi V \left(\frac{2mkT_c}{h^2} \right)^{\frac{3}{2}} \sum_{k=0}^{\infty} \int_0^{\infty} x^{\frac{1}{2}} e^{-(k+1)x} dx = 2\pi V \left(\frac{2mkT_c}{h^2} \right)^{\frac{3}{2}} \Gamma\left(\frac{3}{2}\right) \sum_{k=0}^{\infty} \frac{1}{(k+1)^{\frac{3}{2}}}$$

$$= V \left(\frac{2\pi mkT_c}{h^2} \right)^{\frac{3}{2}} \sum_{k=1}^{\infty} \frac{1}{k^{\frac{3}{2}}} \approx 2.612 V \left(\frac{2\pi mkT_c}{h^2} \right)^{\frac{3}{2}}$$

當 $T < T_c$ 時, 因為 $\mu \leq 0$, 所以 $\mu = 0$. $N_{\epsilon > 0} = 2.612 V \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}}$, $N_0 = N_{\epsilon=0} = N \left[1 - \left(\frac{T}{T_c} \right)^{\frac{3}{2}} \right]$

$$\bar{E} = \int_0^{\infty} \frac{\epsilon g(\epsilon)}{\exp(\frac{\epsilon}{kT}) - 1} d\epsilon = 2\pi V \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^{\infty} \frac{\epsilon^{\frac{3}{2}}}{\exp(\frac{\epsilon}{kT}) - 1} d\epsilon = \frac{2\pi V}{h^3} \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} \int_0^{\infty} \frac{x^{\frac{3}{2}}}{\exp(x) - 1} dx$$

$$= 2\pi V \left(\frac{2mkT}{h^2} \right)^{\frac{3}{2}} \cdot kT \int_0^{\infty} \frac{x^{\frac{3}{2}}}{e^x - 1} dx$$

$$= 2\pi V kT \left(\frac{2mkT}{h^2} \right)^{\frac{3}{2}} \Gamma\left(\frac{5}{2}\right) \sum_{n=1}^{\infty} \frac{1}{n^{\frac{5}{2}}}$$

$$= \frac{3}{2} V kT \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} \sum_{n=1}^{\infty} \frac{1}{n^{\frac{5}{2}}} \approx 1.341 \times \frac{3}{2} V kT \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} = 2.0115 V \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} kT$$

$$C_V = \left(\frac{\partial E}{\partial T} \right)_V = 2.0115 \times \frac{5}{2} V k \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} = 5.0288 V k \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}}$$

(2) ~~光子自旋~~ $\mu = 0$

光子自旋簡並度 $J=2$. $\epsilon = cp = h\nu$

$$\Omega(\epsilon) = J \cdot V \cdot \frac{4}{3} \pi \left(\frac{\epsilon}{c} \right)^3 \Rightarrow g(\epsilon) = \frac{4\pi V}{3h^3} J \times 3 \left(\frac{\epsilon}{c} \right)^2 \cdot \frac{1}{c} = \frac{4\pi V}{(hc)^3} J \epsilon^2 = \frac{8\pi V}{(hc)^3} \epsilon^2$$

因為光子不斷在空腔內產生和消滅, 所以 $\alpha = 0 \Rightarrow \mu = 0$

$$n(\epsilon) d\epsilon = \frac{g(\epsilon)}{e^{\frac{\epsilon}{kT}} - 1} d\epsilon = \frac{8\pi V}{(hc)^3} \cdot \frac{\epsilon^2}{e^{\frac{\epsilon}{kT}} - 1} d\epsilon$$

$$\bar{E} = \int_0^{\infty} \epsilon n(\epsilon) d\epsilon = \frac{8\pi V}{(hc)^3} \int_0^{\infty} \frac{\epsilon^3}{e^{\frac{\epsilon}{kT}} - 1} d\epsilon = \frac{8\pi V}{(hc)^3} (kT)^4 \Gamma(4) \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{8\pi V}{(hc)^3} (kT)^4 \cdot 3! \cdot \frac{\pi^4}{90} = \frac{8\pi^5 k^4}{15 (hc)^3} V T^4$$

习题 2.

$$s = \frac{1}{2}, \quad J = 2s + 1 = 2$$

$$g(\epsilon) = 2\pi V J \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \sqrt{\epsilon} \Rightarrow N = \int_0^{\epsilon_F} g(\epsilon) d\epsilon = \int_0^{\epsilon_F} 2\pi V J \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \sqrt{\epsilon} d\epsilon = \frac{4}{3} \pi V J \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \epsilon_F^{\frac{3}{2}}$$

$$\Rightarrow \epsilon_F = \left(\frac{3N}{4\pi V J}\right)^{\frac{2}{3}} \frac{h^2}{2m} = \cancel{\left(\frac{3N}{8\pi V J}\right)^{\frac{2}{3}} \frac{h^2}{2m}}$$

$$\Rightarrow \bar{E} = \int_0^{\epsilon_F} \epsilon g(\epsilon) d\epsilon = 2\pi V J \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \cdot \frac{2}{5} \epsilon_F^{\frac{5}{2}} = \frac{2}{5} N \cdot \epsilon_F = \frac{2}{5} N \epsilon_F = \frac{2}{5} N \left(\frac{3N}{4\pi V}\right)^{\frac{2}{3}} \frac{h^2}{2m}$$

$$2) \quad P = -\frac{\partial \bar{E}}{\partial V} = -\frac{\partial}{\partial V} (\bar{E} - TS) = -\frac{\partial \bar{E}}{\partial T}$$

$$3) \quad P = -\frac{\partial \bar{E}}{\partial V} = \frac{2}{5} N \left(\frac{3N}{4\pi V}\right)^{\frac{2}{3}} \cdot \frac{1}{V} \cdot \frac{h^2}{2m} = \frac{2}{5} \cdot \frac{N \epsilon_F}{V}$$

$$\Phi(\beta, V) = \int_0^{\infty} g(\epsilon) \ln(1 - e^{-\beta \epsilon}) d\epsilon = \cancel{4\pi V \left(\frac{2m}{h^2}\right)^{\frac{3}{2}}} \int_0^{\infty} \epsilon^{\frac{1}{2}} \ln(1 - e^{-\beta \epsilon}) d\epsilon$$

$$= -\frac{8\pi V}{3} \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \int_0^{\infty} \ln(1 - e^{-\beta \epsilon}) d(\epsilon^{\frac{3}{2}})$$

$$= \frac{8\pi V}{3} \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \int_0^{\infty} \frac{+\beta e^{-\beta \epsilon} \epsilon^{\frac{3}{2}}}{1 - e^{-\beta \epsilon}} d\epsilon = \frac{8\pi V \beta^{\frac{3}{2}}}{3} \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \int_0^{\infty} \frac{x^{\frac{3}{2}} e^{-x}}{1 - e^{-x}} dx$$

$$= \frac{8\pi V}{3} \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \beta^{-\frac{3}{2}} \Gamma\left(\frac{5}{2}\right) \sum_{n=1}^{\infty} \frac{1}{n^{\frac{5}{2}}}$$

$$= 1.341 \times 2V \left(\frac{2m\pi}{h^2}\right)^{\frac{3}{2}} \beta^{-\frac{3}{2}}$$

$$P = \frac{1}{\beta} \frac{\partial \Phi}{\partial V} = 1.341 \times 2V \left(\frac{2m\pi kT}{h^2}\right)^{\frac{3}{2}} \cdot kT$$

$$\Phi(\alpha, \beta, V) = \int_0^{\infty} g(\epsilon) \ln(1 + e^{-\alpha - \beta \epsilon}) d\epsilon$$

$$= 4\pi V \left(\frac{2m}{h^2}\right)^{\frac{3}{2}} \int_0^{\infty} \epsilon^{\frac{1}{2}} \ln(1 + e^{-\alpha - \beta \epsilon}) d\epsilon$$

(4)